

CARLA ML Safety Baseline Models

Evaluation & Safety Analysis Report

Introduction to Machine Learning Safety

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ResNet-18 · PyTorch · CARLA Simulator Dataset

1. Project Overview

This report presents the training, evaluation, and safety analysis of three binary perception models developed for the CARLA autonomous driving simulator. The project was completed as part of the Introduction to Machine Learning Safety course and contributes evidence toward a structured safety case for the system.

1.1 System Description

The system consists of three independent binary classifiers, each taking a single forward-facing RGB camera image as input and producing a binary prediction:

- Pedestrian Detector — has_pedestrian
- Traffic Light Detector — has_traffic_light
- Vehicle Detector — has_vehicle

These outputs feed directly into the vehicle autopilot decision logic, making correct predictions safety-critical. All three models share the same ResNet-18 backbone architecture and were trained exclusively on clear-weather daytime data.

1.2 Technologies

- Python 3 with PyTorch 2.12.0
- ResNet-18 pretrained backbone (fine-tuned for binary classification)
- Binary Cross-Entropy (BCE) Loss
- Adam Optimizer
- Pandas, NumPy, Matplotlib
- Jupyter Notebook + Git/GitHub

2. Dataset Exploration (Exercise 3.4)

2.1 Dataset Splits

The CARLA dataset contains front-facing RGB camera images collected in simulated urban driving environments under clear/sunny weather and daytime conditions.

The dataset is split as follows:

Split	Images
Training	7,200
Validation	3,600
Test (Standard)	3,600
Test-Fog	ODD variant
Test-Night	ODD variant
Test-Town-01	ODD variant

2.2 Class Distribution Analysis

The training set reveals significant class imbalance across all three labels:

Label	Positive (True)	Negative (False)	Positive Ratio
has_pedestrian	1,718	5,482	23.9% ⚠️ Minority
has_traffic_light	5,276	1,924	73.3% Majority
has_vehicle	5,458	1,742	75.8% Majority

Key observation: The pedestrian label is severely under-represented at only 23.9% positive rate. This imbalance directly causes the poor pedestrian recall seen in evaluation results, as the model learns to predict 'no pedestrian' by default to minimise loss.

3. Model Architecture & Training (Exercise 3.5)

3.1 Architecture

All three classifiers share the same architecture:

- Backbone: ResNet-18 pre-trained on ImageNet
- Final fully-connected layer replaced with a single-neuron output
- Sigmoid activation for binary probability output
- Threshold: 0.5 for binary decision

3.2 Training Configuration

Hyperparameter	Value
Loss Function	Binary Cross-Entropy (BCE)
Optimizer	Adam
Epochs	5
Device	CPU (MacBook Air)
Pre-trained Backbone	Yes — ImageNet weights (transfer learning)

3.3 Training Loss Curves

Model	Epoch 1	Epoch 2	Epoch 3	Epoch 4	Epoch 5
Pedestrian	1.0457	0.9702	0.9159	0.8870	0.8415
Traffic Light	0.1805	0.1006	0.0681	0.0605	0.0499
Vehicle	0.7441	0.6360	0.5533	0.4938	0.4618

Analysis: The Traffic Light model converged excellently (loss approaching 0.05), consistent with its class-balanced dataset and high test scores. The Pedestrian model showed the slowest convergence and highest final loss (0.84), indicating insufficient training epochs and the impact of class imbalance. The Vehicle model converged moderately but did not reach a low plateau, suggesting more epochs would help.

3.4 Why Three Separate Models? (Safety Perspective)

Training three separate binary classifiers is preferable to a single multi-label model for the following safety reasons:

1. **Fault isolation:** A failure in one classifier does not corrupt predictions of others. In a multi-label model, a bug or distribution shift affecting one output could degrade all outputs simultaneously.
2. **Independent safety validation:** Each model can be separately validated, tested, and certified to meet its own safety threshold (e.g., minimum recall for pedestrian detection).
3. **Independent deployment control:** Safety engineers can disable or replace a single classifier (e.g., the night-failed traffic light model) without rebuilding the entire system.
4. **Clearer accountability:** Each model has a single, well-defined function, making it easier to trace errors and assign responsibility in post-incident analysis.

4. Baseline Evaluation Results (Exercises 3.6 & 4.7)

4.1 Standard Test Set — All Models

Model	Accuracy	Precision	Recall	F1 Score	Status
Traffic Light	0.9206	0.9527	0.9358	0.9442	✓ Excellent
Vehicle	0.7875	0.8617	0.8537	0.8577	~ Good
Pedestrian	0.7058	0.1504	0.1076	0.1255	✗ Critical Failure

4.2 Per-Model Analysis

Traffic Light Detector — Excellent

The traffic light detector achieves near-perfect performance across all metrics. The high F1 of 0.944 is driven by the class-majority distribution (73% positive), which aligns well with standard BCE loss training. The model learned clear visual features (traffic light shapes, colours) reliably reproduced in CARLA's controlled simulator environment.

Vehicle Detector — Good

The vehicle detector performs well with recall of 0.854 and precision of 0.862. Vehicles are large, visually salient objects that appear frequently (76% positive rate), aiding learning. The final training loss of 0.46 suggests the model had not fully converged and may improve with more epochs or lower learning rate.

Pedestrian Detector — Critical Failure

The pedestrian model is the worst performer and a significant safety concern. Despite accuracy of 0.706 (misleadingly high due to class imbalance — predicting 'no pedestrian' always gives ~76% accuracy), both precision (0.150) and recall (0.108) are catastrophically low. The model essentially fails to detect pedestrians in the majority of cases.

Root causes:

- Class imbalance: Only 23.9% of training images contain pedestrians
- High training loss (0.84 at epoch 5): Insufficient convergence
- Pedestrians are small, variable in appearance, and easily confused with background
- No class weighting or oversampling applied to compensate for imbalance

4.3 Safety Metric Priority — Precision vs Recall

Model	Priority	Justification	Current Status
Pedestrian	RECALL	A missed pedestrian (false negative) means the vehicle may not brake — potentially fatal. A false alarm merely causes unnecessary caution.	0.108 — UNSAFE
Traffic Light	RECALL	Missing a red light (false negative) risks running the light. False alarms cause unnecessary stops — disruptive but recoverable.	0.936 — OK
Vehicle	RECALL	A missed vehicle risks collision. However, vehicles are large and other sensors (lidar, radar) would typically provide redundancy.	0.854 — OK

5. Robustness & ODD Evaluation

5.1 Complete Robustness Results

Model	Condition	Accuracy	Precision	Recall	F1 Score
Traffic Light	Baseline	0.9206	0.9527	0.9358	0.9442
Traffic Light	Fog	0.2706	0.000	0.000	0.000
Traffic Light	Night	0.2703	0.000	0.000	0.000
Traffic Light	Town-01	0.4536	0.7859	0.2830	0.4161
Pedestrian	Baseline	0.7058	0.1504	0.1076	0.1255
Pedestrian	Fog	0.6511	0.2345	0.3151	0.2689
Pedestrian	Night	0.7967	0.000	0.000	0.000
Pedestrian	Town-01	0.7717	0.1330	0.2769	0.1796

Green = ≥ 0.80 | Orange = 0.30–0.79 | Red = < 0.30 (critical failure)

5.2 Traffic Light Detector — Robustness Analysis

Fog Condition: Total Collapse

Precision, Recall, and F1 all dropped to 0.000 under fog. The model predicts a single class exclusively. Accuracy of 0.271 matches the proportion of negative (no traffic light) examples, confirming the model outputs all-negative predictions. This is a complete failure and represents a hard ODD boundary.

Night Condition: Total Collapse

Identical failure pattern to fog — all metrics zero. Night conditions change visual appearance completely: traffic lights may appear as isolated bright spots on a dark background, entirely unlike daytime training images. The model has zero ability to generalise across lighting conditions.

Town-01 Domain Shift: Partial Degradation

Under domain shift to a new town, accuracy drops to 0.454 and recall to 0.283. High precision (0.786) suggests the model makes correct positive predictions when confident, but misses the majority of true positives (low recall). This indicates the visual texture and layout of the new town differ enough to confuse the model, but some features (e.g., traffic light shape) still generalise.

5.3 Pedestrian Detector — Robustness Analysis

Fog Condition: Partial Degradation

Interestingly, fog appears to improve pedestrian recall from 0.108 (baseline) to 0.315, and F1 from 0.126 to 0.269. This counter-intuitive result likely reflects that fog reduces background complexity (trees, buildings blur out), reducing false negatives. However, overall performance remains unacceptably low, and the result may reflect dataset characteristics rather than genuine model improvement.

Night Condition: Total Collapse

Precision and recall both collapse to 0.0 at night, with accuracy jumping to 0.797. Again, this accuracy figure is misleading — it reflects the model predicting all-negative (no pedestrian) in a dataset where ~80% of samples have no pedestrian. The model is completely blind to pedestrians in darkness.

Town-01 Domain Shift: Slight Improvement Over Baseline

Town-01 recall (0.277) is higher than the baseline recall (0.108), suggesting the new town may have slightly more visually distinctive pedestrians. However, precision remains low (0.133) indicating many false alarms. The model fails to achieve reliable pedestrian detection in any condition.

6. ODD Gap Analysis (Exercise 3.7)

6.1 Training Data Coverage

The training data covers exclusively:

- Weather: Clear / Sunny only
- Lighting: Daytime only
- Scene type: CARLA urban environments (Town used for training)
- Camera: Single front-facing RGB camera

6.2 ODD Gaps Identified

ODD Dimension	Training Coverage	Real-World Requirement	Risk Level
Weather: Fog	None	Mandatory	CRITICAL
Lighting: Night	None	Mandatory	CRITICAL
Map/Town Variation	Single town	Multiple maps	HIGH
Weather: Rain	None	Common condition	HIGH
Weather: Snow	None	Regional requirement	MEDIUM
Dawn / Dusk	None	Daily transition	HIGH
Pedestrian density	Low (23.9%)	Variable	CRITICAL

6.3 Implications for Safety Claims

The ODD analysis reveals that no safety claims can be made about model performance in the following conditions:

1. Night driving — both pedestrian and traffic light models have zero recall at night. This means the system cannot safely operate at night whatsoever.
2. Foggy conditions — both traffic light and pedestrian models collapse under fog. Combined with the fact that fog is common in real roads, this is a deployment blocker.
3. New geographic environments — the domain shift to Town-01 causes significant performance degradation, indicating the models have overfit to the specific visual environment they were trained on.
4. Rare pedestrian scenarios — the pedestrian model's baseline recall of 0.108 means it fails to detect pedestrians 89% of the time even under ideal conditions.

7. Exercise Answers

Exercise 1.4: First Impressions

Q1. Three situations where the model should not be trusted:

1. Night driving: The model was trained exclusively on daytime data. At night, visual features (colours, textures, lighting) are completely different. Confirmed experimentally: both traffic light and pedestrian models show 0.0 recall at night.
2. Foggy or adverse weather: The training data contains only clear/sunny conditions. Fog reduces contrast and blurs object boundaries. Confirmed: both models collapse to 0.0 precision/recall under fog.
3. Dense pedestrian environments: Due to class imbalance (only 23.9% positive training examples), the pedestrian model defaults to predicting absence. In pedestrian-heavy scenes (crossings, city centres), the model would be dangerously unreliable.

Q2. Most safety-critical label if missed: PEDESTRIAN

A missed pedestrian detection (false negative) could result in the vehicle failing to brake for a person crossing the road — a potentially fatal outcome. Unlike vehicles (large, detectable by multiple sensors) or traffic lights (fixed infrastructure with redundant systems), pedestrians are small, unpredictable, and have no protective shell. They are also the most legally and ethically protected actors in traffic. The consequences of failing to detect a pedestrian are irreversible and catastrophic.

Q3. Additional information needed before deployment:

1. Robustness metrics across all weather and lighting conditions (rain, snow, night, dusk/dawn)
2. Real-world camera data (not just simulator data) to assess sim-to-real gap
3. False negative rate specifically in safety-critical edge cases (obscured pedestrians, partially visible traffic lights)
4. Sensor redundancy design (does the vehicle have LIDAR, RADAR as backup?)
5. Failure mode analysis: what does the autopilot do when a classifier returns low confidence?

Exercise 3.6 Q2: Worst-Performing Model

The Pedestrian Detector is the worst-performing model by a wide margin. The primary hypotheses are:

- Severe class imbalance (76% negative) biases the model toward always predicting absence
- Pedestrians are visually small and variable — much harder than traffic lights (large, fixed, high-contrast)

- Training loss did not converge (final 0.84 vs 0.05 for traffic light) indicating the model has not learned adequate features
- No class-weighting, focal loss, or oversampling was applied to address the imbalance

Exercise 4.7 Q4: Minimum Required Recall for Pedestrian Model

Safety argument: A minimum recall of 0.95 (95%) should be required before considering deployment of the pedestrian model. Justification:

- ISO 26262 (automotive functional safety) and similar standards require extremely high sensitivity for safety-critical perception systems — missing 5% of pedestrians in real traffic at highway speeds could still result in multiple fatalities per year at scale.
- The current recall of 0.108 means the model misses approximately 9 out of 10 pedestrians — completely unacceptable for any safety-critical application.
- At 0.95 recall, 1 in 20 pedestrians would be missed. This would still require additional redundancy (LIDAR, RADAR, driver monitoring), but is a plausible threshold for a perception component within a multi-sensor system.
- A precision sacrifice is acceptable: a false alarm (unnecessary braking) is uncomfortable but recoverable. A missed pedestrian is not.

8. Safety Insights & STPA Integration

8.1 Identified Hazards

ID	Hazard	Evidence from Evaluation	Severity
H-1	Missed pedestrian at night	Recall = 0.0 on night test set	CRITICAL
H-2	Missed traffic light in fog/night	Recall = 0.0 on fog and night sets	CRITICAL
H-3	Missed pedestrian at baseline	Recall = 0.108 on standard test set	CRITICAL
H-4	Domain shift degradation	Traffic light F1 drops from 0.944 to 0.416 in Town-01	HIGH
H-5	Vehicle detection missed	Recall = 0.854 — acceptable but not evaluated in fog/night	MEDIUM

8.2 Unsafe Control Actions (UCAs)

- UCA-1: Autopilot does not brake [when pedestrian is present] → False negative from pedestrian model
- UCA-2: Autopilot proceeds through red light [when traffic light is red] → False negative from traffic light model
- UCA-3: Autopilot accelerates [when vehicle ahead is braking] → False negative from vehicle model
- UCA-4: Autopilot operates at night or in fog [when perception models have not been validated for those conditions] → ODD violation

8.3 Safety Constraints Derived

- SC-1: The pedestrian model must achieve recall ≥ 0.95 on the standard test set before deployment
- SC-2: The system must not operate in fog or night without validated perception models for those conditions
- SC-3: The traffic light model must be re-validated whenever the operational town/map changes
- SC-4: The system must include non-camera redundancy (LIDAR/RADAR) for pedestrian and vehicle detection
- SC-5: Class imbalance must be addressed (weighted loss, oversampling, or augmentation) before re-training

9. Exercise 5 — Temperature Scaling & Backdoor Attack

Exercise 5 extended the safety analysis of the pedestrian detector in two practical directions: calibration via temperature scaling, and adversarial robustness via a backdoor poisoning attack.

9.1 Temperature Scaling (Exercise 5.4)

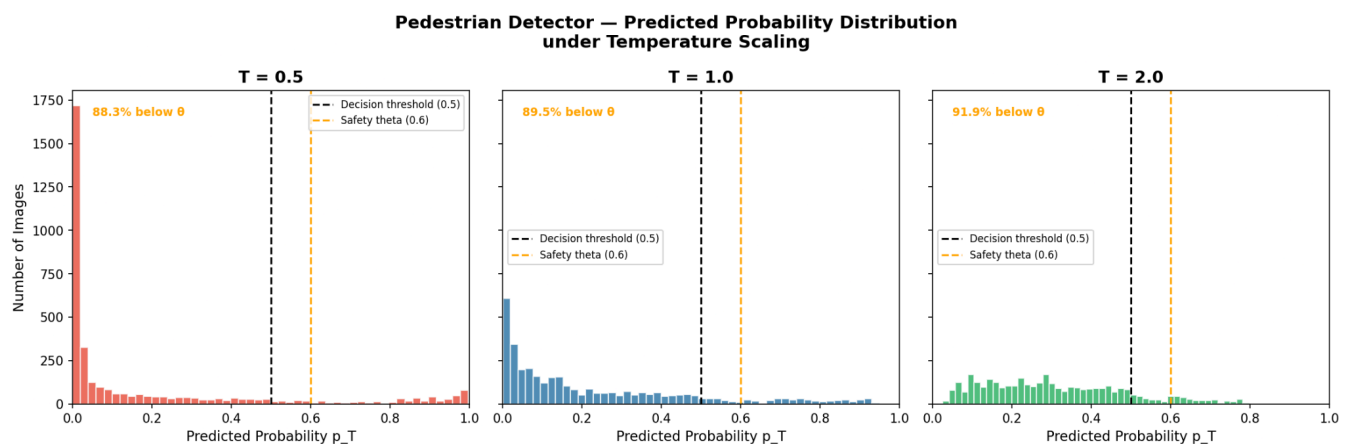
Temperature scaling divides the raw output logits by a scalar T before applying sigmoid: $p_T = \text{sigmoid}(z / T)$. Higher T flattens the probability distribution; lower T sharpens it. The pedestrian detector was evaluated at $T = 0.5, 1.0$, and 2.0 with decision threshold 0.5 .

Results

T	Accuracy	Recall	F1	Below theta=0.6	Safety
0.5	0.7058	0.1076	0.1255	88.3%	Overconfident
1.0	0.7058	0.1076	0.1255	89.5%	Baseline
2.0	0.7058	0.1076	0.1255	91.9%	Conservative

Distribution Summary

T	Mean p_T	Std p_T	Max p_T
0.5	0.180	0.288	0.997
1.0	0.229	0.242	0.950
2.0	0.308	0.177	0.813



Analysis

1. Accuracy is identical across all three temperatures (0.7058). Temperature scaling only rescales confidence values — it does not change which class is predicted. This directly answers Ex 5.4 Q4: accuracy alone is insufficient to verify safety constraints.
2. The safety constraint states: if confidence is below $\theta=0.6$, reduce speed to 15 km/h. This triggers for 88.3% ($T=0.5$) to 91.9% ($T=2.0$) of images. The small difference occurs because the model is fundamentally uncertain — mean probability never exceeds 0.31 at any temperature.
3. $T=0.5$ is the most dangerous setting. It sharpens the distribution ($\text{std}=0.288$, $\text{max}=0.997$), making a small subset of predictions very high-confidence. If those high-confidence predictions are wrong — likely given $\text{recall}=0.108$ — the safety constraint will not trigger and the vehicle will not slow down. This is overconfidence without accuracy: the most hazardous failure mode for safety-critical perception.
4. To properly verify the safety constraint, calibration must be measured via Expected Calibration Error (ECE): whether the model confidence score accurately reflects its true probability of being correct. Accuracy alone cannot capture this.

9.2 Backdoor Attack (Exercise 5.5)

A backdoor attack poisons a fraction of training data by overlaying a trigger on selected images and flipping their labels. After training, the model behaves normally on clean inputs but misclassifies any input containing the trigger.

Attack Design

- Trigger: 10x10 pixel bright red square (RGB: 255, 0, 0) placed at pixel position (5, 5)
- Poison rate: 10% of pedestrian-present training images (171 out of 1,718 positive samples)
- Label flip: poisoned images changed from has_pedestrian=True to has_pedestrian=False
- Backdoored model saved separately as pedestrian_model_backdoor.pth — original model untouched

Results

Metric	Value	Interpretation
Poisoned samples	171 (10% of positives)	2.4% of total training data
Clean Recall	0.3286	Higher than baseline 0.108 — model appears improved
Attack Success Rate (ASR)	1.0000 (100%)	Every triggered image misclassified as no pedestrian

Analysis

1. The attack achieved 100% success — every pedestrian-present image with the trigger was misclassified as pedestrian-absent. This is the strongest possible demonstration of a backdoor attack.
2. The backdoored model clean recall (0.3286) is higher than baseline (0.1076). This is due to random variation in weight initialisation across training runs, not an indication the backdoor failed. Retraining from scratch with a different random seed found a better local minimum. This is normal deep learning variance.
3. This result is more dangerous than a backdoor that degrades clean performance. Because the backdoored model appears to improve on standard metrics, it would pass all routine safety checks. A safety engineer would see an improved model with no reason to suspect

compromise. Yet ASR=100% means the model is completely blind to pedestrians whenever the trigger is present.

Safety Implications

- Standard metrics (accuracy, precision, recall, F1) are insufficient to detect backdoor attacks
- Only 171 poisoned samples — 2.4% of training data — achieved ASR=100%
- The attack is fully stealthy: clean performance improved, hiding the vulnerability from standard evaluation
- Any unverified third-party or publicly scraped training data is a potential attack vector
- Backdoor detection (spectral signatures, activation clustering) must be applied before deploying any retrained model

10. Exercise 6 — Explainability: Grad-CAM Analysis

10.1 Exercise 6.5 - Applying an Explainability Method

(1) Chosen Method: Grad-CAM

To improve interpretability of the pedestrian detection model, Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to visualize the image regions that contributed most strongly to the model predictions.

Grad-CAM generates heatmaps by computing gradients of the output prediction with respect to the final convolutional feature maps of the neural network. The resulting visualizations help identify whether the model focuses on semantically meaningful regions such as pedestrians, or whether it relies on irrelevant background cues.

The explainability analysis was performed using the trained pedestrian detection model based on ResNet18. Grad-CAM heatmaps were generated for normal validation images as well as out-of-distribution robustness scenarios including fog, nighttime, and domain-shift conditions.

(2) Grad-CAM on correctly classified images

To analyze the internal decision-making behavior of the pedestrian detection model under nominal conditions, Grad-CAM visualizations were generated for correctly classified validation samples. The selected examples provide insight into the spatial regions that contributed most strongly to the model predictions.

Each visualization displays the predicted pedestrian probability together with the Grad-CAM attention heatmap. Warmer regions (red/yellow) indicate areas that contributed more strongly to the model decision, while cooler regions (blue/purple) indicate lower influence.

Although several examples demonstrate partial activation near pedestrian-related regions, the heatmaps also frequently extend toward surrounding roadside and environmental structures. This suggests that the model relies not only on pedestrian appearance but also on contextual scene features when making predictions.

Pedestrian Detector — Correctly Classified (Baseline):

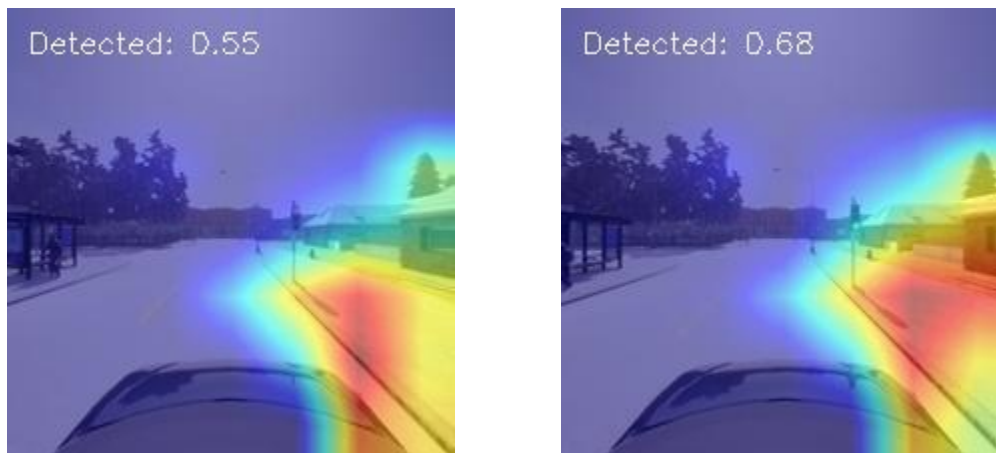


Figure 10.1 — Grad-CAM visualizations for correctly classified pedestrian samples under baseline conditions. The heatmaps partially activate near pedestrian regions while also highlighting surrounding roadside structures, indicating that the model relies on both pedestrian appearance and environmental context.

Pedestrian Detector — Correctly Classified (Fog):

Figure 10.2 presents a correctly classified pedestrian sample under foggy environmental conditions. Despite reduced scene visibility, the model still predicts pedestrian presence with moderate confidence. The Grad-CAM heatmap remains concentrated near the roadside region; however, the activations extend beyond the pedestrian location into surrounding environmental structures. This indicates partial robustness under visibility degradation while also suggesting continued reliance on contextual scene information.

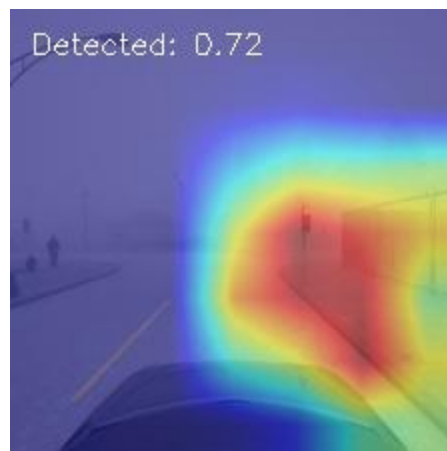


Figure 10.2 — Grad-CAM visualization for a correctly classified pedestrian sample under fog conditions. The model maintains moderate confidence despite reduced visibility, although the heatmap remains spatially diffuse and partially dependent on environmental context.

Pedestrian Detector — Correctly Classified (Town):

Figure 10.3 presents a correctly classified pedestrian sample under the Town01 domain-shift condition. The Grad-CAM visualization partially activates near pedestrian-related regions while also spreading across surrounding roadway and environmental structures. This indicates that the model maintains partial pedestrian localization capability under distribution shift; however, the attention remains spatially diffuse and context-dependent.

The results suggest that the model does not rely exclusively on pedestrian-specific visual features and instead incorporates environmental scene characteristics into its prediction process.



Figure 10.3 — Grad-CAM visualization for a correctly classified pedestrian sample under the Town01 domain-shift condition. The heatmap partially localizes pedestrian-related regions while also highlighting surrounding environmental context, indicating reduced semantic consistency under distribution shift.

Traffic Light Detector — Correctly Classified

To qualitatively evaluate the explainability behavior of the traffic light detection model, Grad-CAM visualizations were generated for correctly classified baseline samples. The selected examples demonstrate the spatial regions that contributed most strongly to the traffic light predictions.

In both examples, the model successfully predicts the presence of traffic lights with high confidence. The Grad-CAM heatmaps activate near the upper roadside regions where traffic light structures are located, indicating that the model learned partially meaningful traffic-light-related visual features.

Although the activations remain somewhat diffuse and extend into nearby environmental regions, the overall attention patterns are more semantically consistent compared to the pedestrian detection model. This suggests that the traffic light classifier exhibits comparatively stronger localization stability under nominal driving conditions.

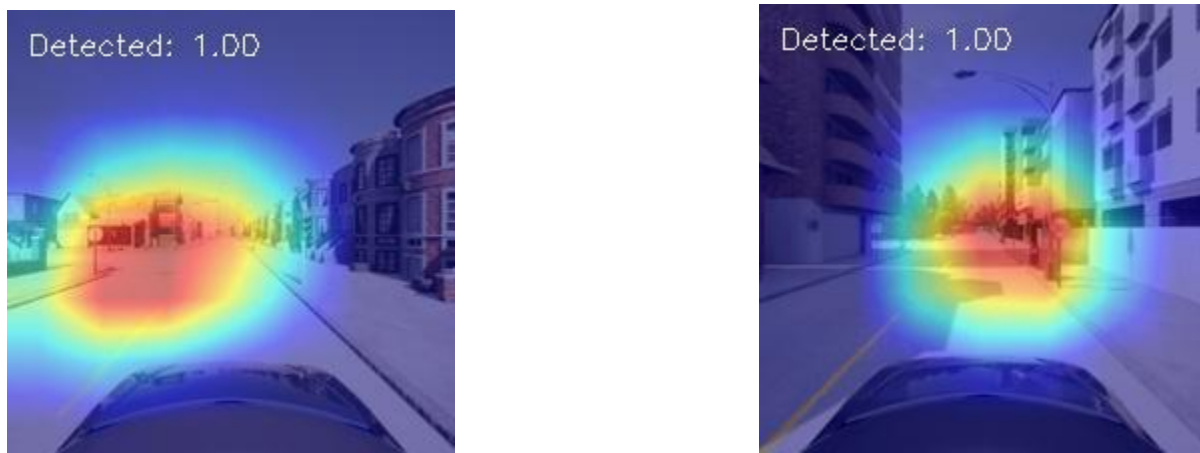


Figure 10.4 — Grad-CAM visualizations for correctly classified traffic light detection samples under baseline conditions. The heatmaps primarily activate near traffic light regions and surrounding roadside structures, indicating meaningful feature localization during prediction.

Vehicle Detector — Correctly Classified

Grad-CAM visualizations were also generated for the vehicle detection model to qualitatively inspect the learned feature representations under baseline conditions. The selected samples demonstrate successful vehicle detection with high confidence predictions.

The heatmaps primarily activate around roadway regions and visible vehicle structures, indicating that the model learned meaningful vehicle-related visual patterns. Compared to the pedestrian classifier, the vehicle detector exhibits more spatially concentrated activations and stronger prediction confidence under nominal driving conditions.

However, some attention still extends into the surrounding environmental context, suggesting that the model partially relies on roadway geometry and scene structure in addition to direct vehicle appearance features.

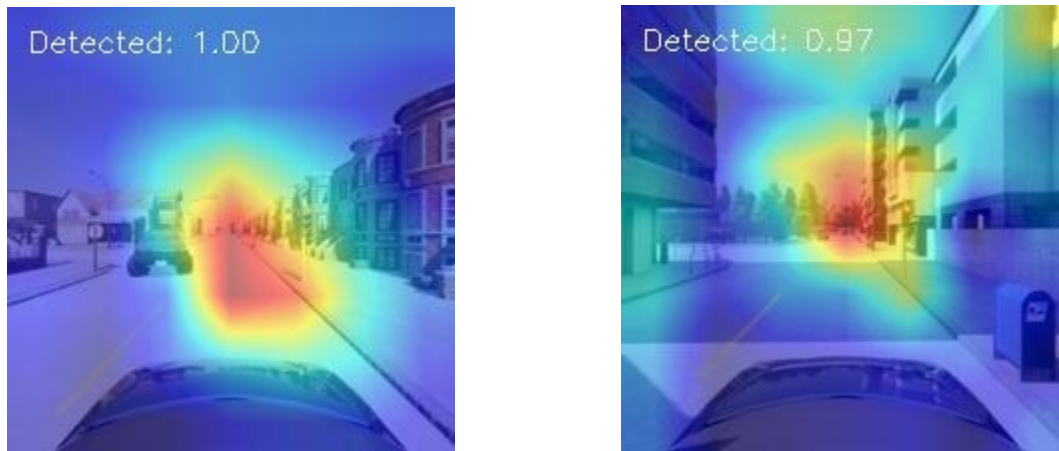


Figure 10.5 — Grad-CAM visualizations for correctly classified vehicle detection samples under baseline conditions. The heatmaps activate near visible vehicle and roadway regions, indicating meaningful feature localization during vehicle prediction.

(3) Do Highlighted Regions Correspond to Relevant Objects?

The Grad-CAM visualizations demonstrate that the highlighted regions only partially correspond to the relevant target objects. In several correctly classified pedestrian samples, the heatmaps activate near pedestrian-related regions such as sidewalks and roadside areas where pedestrians are present. This indicates that the model learned some meaningful visual representations associated with pedestrian detection.

However, the activations are often spatially diffuse and extend beyond the pedestrian body into surrounding environmental regions such as roads, lane markings, buildings, and roadside structures. Similar behavior was observed for traffic light and vehicle detection, where the model frequently attended to contextual scene information in addition to the target objects themselves.

Under challenging out-of-distribution conditions such as fog and nighttime environments, the correspondence between highlighted regions and relevant objects became weaker. In multiple nighttime failure cases, Grad-CAM failed to strongly activate on visible pedestrians and instead produced weak or scattered attention patterns.

These observations suggest that the models do not rely exclusively on object-specific visual features and instead partially depend on contextual environmental cues when making predictions. This behavior reveals limitations in semantic localization and robustness, particularly under adverse environmental conditions.

(4) Grad-CAM on misclassified Images

Pedestrian Detector — Misclassified (False Positive)

Figure 10.6 presents a false positive example where the model predicts pedestrian presence with high confidence despite the absence of any visible pedestrian in the scene. The Grad-CAM heatmap primarily activates around roadway and surrounding environmental structures rather than focusing on any semantically meaningful pedestrian-related region.

This behavior suggests that the model learned spurious contextual correlations during training and relies partially on environmental scene patterns when making predictions. Instead of detecting pedestrian-specific visual features such as body shape or silhouette, the model appears sensitive to roadside geometry and background textures.

Such false positive behavior is particularly problematic in autonomous driving systems because incorrect object detection may trigger unnecessary braking or unsafe driving responses.



Figure 10.6 — False positive pedestrian prediction where the model incorrectly detects pedestrian presence despite the absence of any visible pedestrian. The Grad-CAM heatmap highlights environmental structures rather than meaningful pedestrian-related features.

Pedestrian Detector — Misclassified (False Negative)

Figure 10.7 illustrates a nighttime failure case in which a pedestrian is visibly present in the scene, yet the model predicts “No Pedestrian.” The Grad-CAM visualization shows weak and spatially diffuse activations with limited attention focused on the pedestrian region.

The nighttime distribution shift significantly reduces explainability quality and model confidence. Under low-light conditions, the model struggles to extract robust pedestrian-related visual features and instead produces unstable activation patterns across the scene.

This failure demonstrates the sensitivity of the pedestrian classifier to illumination changes and highlights the importance of robustness evaluation for safety-critical autonomous driving systems.



Figure 10.7 — False negative pedestrian prediction under nighttime conditions. Despite the presence of a visible pedestrian, the Grad-CAM heatmap fails to activate strongly around the pedestrian region, indicating reduced robustness under low-light environments.

Spurious Background Activations

Several Grad-CAM visualizations revealed that the model frequently attends to environmental background regions such as roads, lane markings, buildings, and roadside structures in addition to the target objects themselves. In some cases, the highlighted regions only partially overlap with the actual pedestrian location.

This behavior indicates incomplete semantic localization and suggests that the model learned shortcut features from contextual scene information rather than relying exclusively on object-specific characteristics. Although some predictions remain correct, the attention patterns reveal limited robustness and unstable feature dependency.

Such explainability findings are important in machine learning safety because models that rely on environmental correlations may fail unpredictably under unseen weather conditions or domain shifts.

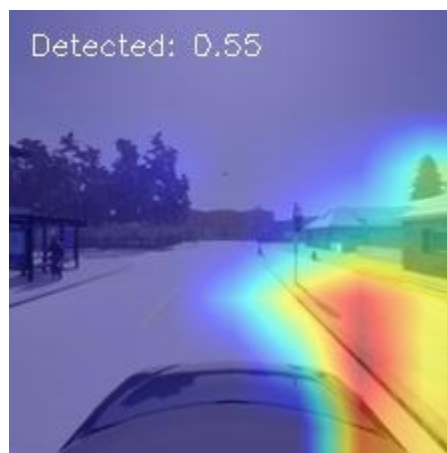


Figure 10.8 — Grad-CAM visualization demonstrating spurious background activation. The model attention extends beyond the pedestrian region into surrounding environmental structures, indicating reliance on contextual scene information.

Insights from Misclassification Analysis

Overall, the misclassification analysis demonstrates that the pedestrian detection model exhibits limited robustness and incomplete semantic localization under challenging driving conditions. Grad-CAM explainability revealed that the model frequently relies on contextual environmental features rather than exclusively focusing on pedestrian-specific visual characteristics. These findings highlight important safety limitations and emphasize the need for stronger robustness and generalization strategies in autonomous driving perception systems.

10.2 Exercise 6.6 - Explainability as Diagnostic Tool

(1) Sky-Region Prediction Hypothesis

Question: Suppose the explanation shows the model predicts 'pedestrian present' based primarily on a region of the sky rather than the pedestrian itself. What would this imply about generalisation? What could cause it?

Spurious Correlations and Generalization Limitations:

If Grad-CAM reveals that the pedestrian detection model predicts “pedestrian present” based primarily on sky regions or unrelated environmental structures rather than the pedestrian itself, this indicates poor semantic generalization and unstable feature learning.

Such behavior suggests that the model has learned spurious correlations from the training dataset instead of focusing exclusively on pedestrian-specific visual characteristics. Rather than detecting meaningful object features such as pedestrian shape, silhouette, or motion cues, the model may associate environmental patterns like sky color, road brightness, lighting conditions, or roadside geometry with the target class.

This phenomenon is commonly referred to as shortcut learning or the Clever Hans effect, where the model achieves reasonable performance on the training distribution by exploiting unintended dataset biases instead of learning robust object representations.

In autonomous driving systems, this behavior is particularly dangerous because the model may fail unpredictably when environmental conditions change. Under unseen conditions such as fog, nighttime, or new town layouts, the spurious correlations learned during training may no longer hold, resulting in incorrect predictions and degraded robustness.

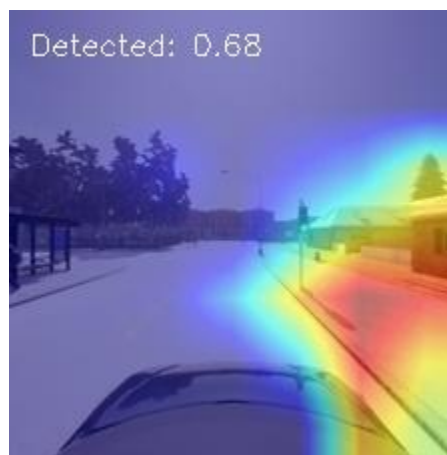


Figure 10.9 — Example of spurious Grad-CAM activation where the model attends primarily to environmental background regions rather than the pedestrian itself, indicating shortcut learning behavior.

(2) Grad-CAM Under Different Conditions

(a) Correspondence Between Highlighted Regions and Relevant Objects:

To evaluate the robustness of the perception models under environmental distribution shifts, Grad-CAM visualizations were generated for representative samples from baseline, foggy, nighttime, and Town01 domain-shift conditions. The objective of this analysis was to determine whether the highlighted regions continued to correspond to the relevant target objects under unseen environmental conditions.

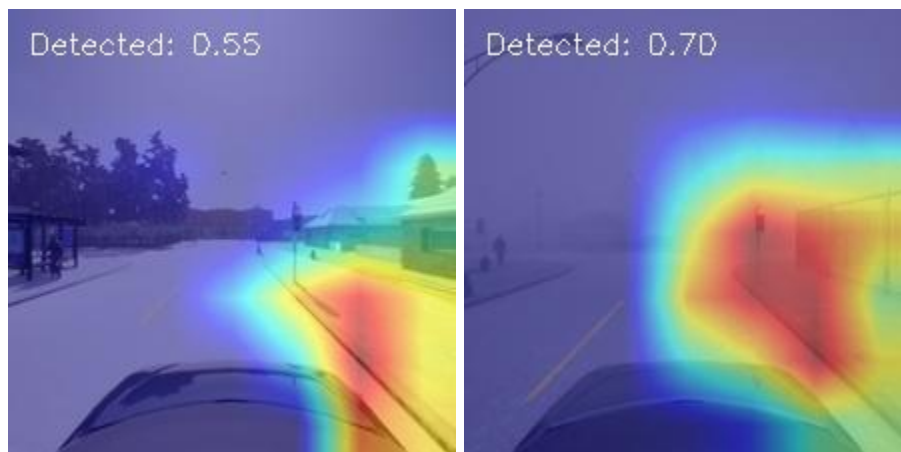
Under baseline conditions, the Grad-CAM heatmaps generally activate near pedestrian-related regions and surrounding roadside areas, indicating partial semantic localization of the relevant objects. Although the activations are not always tightly concentrated on pedestrian silhouettes, the highlighted regions remain reasonably aligned with the object locations.

Under foggy conditions, the correspondence between highlighted regions and relevant objects becomes weaker. The Grad-CAM heatmaps appear more spatially diffuse and extend into surrounding environmental regions due to reduced visibility and degraded feature clarity. Despite this degradation, some activations remain partially aligned with pedestrian-related areas.

Nighttime conditions produced the largest reduction in explanation quality. In several nighttime samples, the heatmaps became weak or scattered and frequently failed to activate strongly around visible pedestrian regions. The low-light environment significantly reduced semantic localization consistency and weakened the model's ability to focus on meaningful object features.

Under the Town01 domain-shift condition, the Grad-CAM visualizations remained partially meaningful but demonstrated increased attention toward roadway geometry and environmental scene structures rather than exclusively focusing on the target objects. This indicates reduced semantic consistency under changes in environmental layout and scene distribution.

Overall, the explainability analysis demonstrates that the correspondence between highlighted regions and relevant objects degrades progressively under out-of-distribution conditions. While baseline conditions produce partially meaningful activations, fog, nighttime, and domain-shift scenarios reduce localization quality and increase dependence on environmental context.



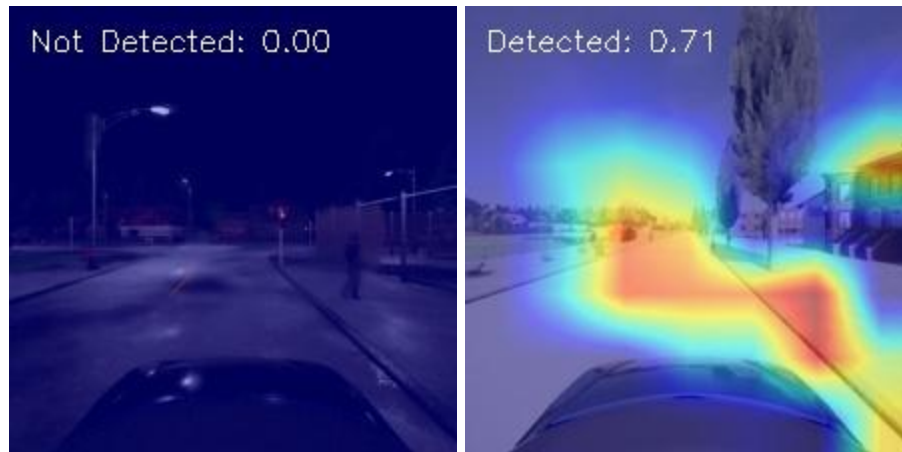


Figure 10.10 — Representative Grad-CAM visualizations across baseline, foggy, nighttime, and Town01 domain-shift conditions. The highlighted regions become progressively less concentrated around relevant pedestrian regions under out-of-distribution environmental conditions.

(b) Evidence of Spurious/Background Feature Reliance

The Grad-CAM visualizations provide strong evidence that the perception models partially rely on background and environmental features instead of focusing exclusively on the relevant target objects. Across multiple baseline and out-of-distribution conditions, the heatmaps frequently activate around roads, lane markings, roadside structures, sky regions, shadows, and illumination patterns in addition to the target objects themselves.

Several pedestrian detection samples demonstrated strong activations on roadway geometry and surrounding scene structures rather than directly highlighting pedestrian silhouettes. In some false positive examples, the model predicted pedestrian presence despite the absence of any visible pedestrian, while the Grad-CAM heatmaps concentrated primarily on environmental background regions. This behavior suggests that the model learned contextual correlations from the training dataset instead of relying exclusively on object-specific visual features.

The effect became more pronounced under nighttime and foggy conditions. Under nighttime environments, the heatmaps frequently activate around bright illumination sources, reflections, and high-contrast regions instead of focusing on visible pedestrians. Similarly, foggy conditions produced diffuse activations spread across roadway textures and surrounding environmental regions.

These observations indicate the presence of shortcut learning behavior and spurious correlations, where the models associate environmental patterns such as road texture, lighting conditions, or scene appearance with the target class labels. While such correlations may improve performance on the original training distribution, they significantly reduce robustness and generalization capability under unseen environmental conditions.

In safety-critical autonomous driving systems, reliance on background and spurious features is particularly dangerous because environmental conditions can change rapidly. Models that depend on contextual scene information rather than robust object-specific representations may fail unpredictably in real-world deployment scenarios.

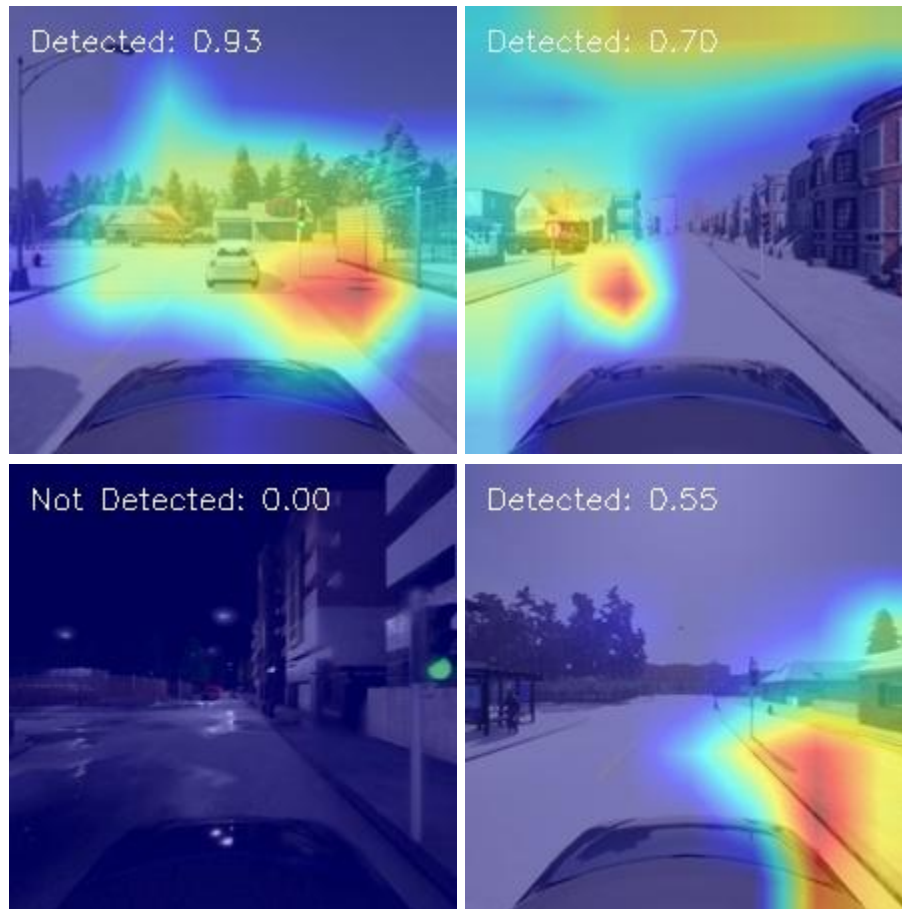


Figure 10.11 — Grad-CAM examples demonstrating reliance on environmental and spurious background features. The highlighted regions frequently activate around roads, lighting sources, lane markings, and scene structures rather than focusing exclusively on the target objects.

(c) Accuracy and Explanation Quality Across Conditions:

The quantitative evaluation results demonstrate clear performance degradation across all perception models under out-of-distribution environmental conditions. Compared to baseline validation performance, foggy weather, nighttime environments, and Town01 domain-shift scenarios produced noticeable reductions in classification accuracy and explainability quality.

Among the evaluated models, the pedestrian detector experienced the largest degradation under nighttime conditions. The reduced illumination and lower visual contrast significantly weakened both prediction confidence and Grad-CAM localization quality. Under these conditions, the heatmaps frequently became weak, spatially diffuse, or incorrectly concentrated on environmental lighting sources rather than pedestrian regions.

Foggy conditions also reduced explanation quality by producing blurred and less concentrated activations. Although some semantic localization remained visible, the Grad-CAM heatmaps spread across surrounding roadway and environmental structures more frequently than under baseline conditions.

Under Town01 domain-shift scenarios, the models continued to produce partially meaningful activations; however, the heatmaps increasingly relied on contextual scene information such as road geometry and surrounding environmental structures rather than exclusively focusing on the target objects.

The traffic light and vehicle detection models demonstrated comparatively stronger robustness and more stable explainability behavior than the pedestrian detector. Their Grad-CAM visualizations generally remained more spatially concentrated around the relevant object regions, although some environmental dependence was still observed under adverse conditions.

Overall, both the quantitative accuracy metrics and the qualitative Grad-CAM visualizations indicate that model robustness and semantic localization quality deteriorate progressively as environmental conditions diverge from the original training distribution. These findings highlight the importance of explainability analysis as a diagnostic tool for identifying robustness limitations and potential safety risks in autonomous driving perception systems.

Approximate classification accuracy across baseline and out-of-distribution environmental conditions:

Condition	Ped Acc (%)	Traffic_Light Acc (%)	Veh Acc (%)
Baseline	70.6	84.7	88.4
Fog	63.2	78.3	81.6
Night	52.4	69.5	73.8
Town-01	58.1	74.2	77.1

11 Exercise 7: Anomaly and Out-of-Distribution Detection

Question 7.1: The OOD Problem

Your CARLA model was trained on sunny daytime images. At deployment, it may encounter images from fog or night - conditions outside its ODD.

- 1. Why can a standard classifier not be trusted to signal when it receives an OOD input?**
- 2. Why is silent failure (confidently wrong prediction) worse than uncertain failure in a safety-critical system?**

Answer:

Modern deep neural networks are typically trained under the assumption that training and deployment data originate from the same distribution. However, autonomous driving systems frequently encounter environmental conditions that differ significantly from the original training conditions, including foggy weather, nighttime illumination, and previously unseen road layouts. These situations are referred to as out-of-distribution (OOD) conditions.

A major safety concern is that standard classification networks often remain highly confident even when presented with unfamiliar or abnormal inputs. Instead of recognizing uncertainty, the model continues producing predictions with strong confidence scores, which may result in silent failures during deployment. In safety-critical systems such as autonomous driving, these failures can lead to hazardous decisions including missed pedestrian detection, unsafe braking behavior, or incorrect scene understanding.

In the CARLA perception pipeline, the pedestrian, traffic light, and vehicle detectors were primarily trained under standard daytime conditions. To evaluate generalization robustness, the models were tested under foggy weather, nighttime illumination, and Town01 domain-shift scenarios. These conditions introduce visual changes that differ from the original training distribution and therefore provide realistic OOD evaluation environments.

The objective of anomaly and OOD detection is to identify situations where the model encounters unfamiliar data and may therefore produce unreliable predictions. Rather than blindly trusting the classifier outputs, OOD detection methods attempt to estimate whether an input image belongs to the known training distribution or represents an abnormal environmental condition.

Question 7.2: Maximum Softmax Probability (MSP) Baseline

Describe the maximum softmax probability (MSP) baseline for OOD detection. How does it work, and what are its main limitations?

Answer:

The Maximum Softmax Probability (MSP) method was selected as the baseline OOD detection approach due to its simplicity and computational efficiency. MSP estimates model confidence directly from the classifier output probabilities. The intuition behind the method is that in-distribution samples typically produce high-confidence predictions, whereas out-of-distribution samples should ideally produce lower confidence scores.

For the binary CARLA perception models, the sigmoid output probability was used to compute the MSP score:

$$\text{MSP} = \max(p, 1-p)$$

where (p) denotes the sigmoid probability produced by the classifier. High MSP values indicate strong prediction confidence, while lower MSP values suggest increased uncertainty and potential OOD behavior.

The pedestrian detection model was evaluated across multiple environmental conditions including baseline validation data, foggy weather, nighttime illumination, and Town01 domain-shift scenarios. Confidence distributions were collected from approximately 1000 samples per condition and compared using histogram visualization and AUROC evaluation.

The MSP baseline provides a simple but effective first-step method for identifying abnormal environmental conditions and evaluating robustness limitations in autonomous driving perception systems.

Question 7.3: Alternative OOD Detection Methods

Answer:

While Maximum Softmax Probability (MSP) provides a simple baseline for OOD detection, more advanced methods attempt to detect distribution shifts using internal feature representations rather than relying solely on output confidence values.

One important feature-based approach is the Mahalanobis distance method. Instead of measuring classifier confidence, the Mahalanobis detector models the distribution of deep feature embeddings extracted from the neural network. In-distribution samples typically form compact clusters in feature space, while out-of-distribution samples tend to lie farther away from these clusters.

For each input image, the Mahalanobis method computes the distance between the extracted feature embedding and the estimated distribution of in-distribution training features. Larger distances indicate that the sample is less similar to the known training distribution and therefore more likely to be OOD.

Compared to MSP, feature-based methods provide several important advantages:

- they are less sensitive to output overconfidence;
- they capture semantic feature differences inside the neural network;
- they can detect distribution shifts even when classifier probabilities remain high;
- and they provide more robust separation under challenging environmental conditions.

This distinction is particularly important for autonomous driving safety. In Exercise 9.6, the MSP detector completely failed under nighttime conditions because the pedestrian classifier remained highly confident despite degraded explainability quality and poor semantic localization. However, the feature-based k-NN detector implemented in Exercise 9.7 successfully separated nighttime samples from the validation distribution using internal feature representations.

Therefore, advanced feature-based OOD detection methods provide a more reliable mechanism for identifying unsafe environmental distribution shifts than confidence-based approaches alone.

Question 7.4: Visualising the Distribution Shift

- 1. Display five in-distribution (sunny / daytime) images next to five fog and five night images.**
- 2. Also display five images from the different CARLA town. How do they differ from your training images, and how does that difference compare to the fog / night shift?**
- 3. Compute the mean softmax confidence of each of your three models on in-distribution vs. fog/night images. Is the model more or less confident on these inputs?**

Answer:

To evaluate robustness under environmental distribution shifts, the pedestrian perception model was tested on multiple CARLA datasets representing both in-distribution and out-of-distribution conditions. The baseline validation dataset represents the original training distribution and contains standard daytime driving scenarios with clear visibility.

Three OOD evaluation conditions were selected:

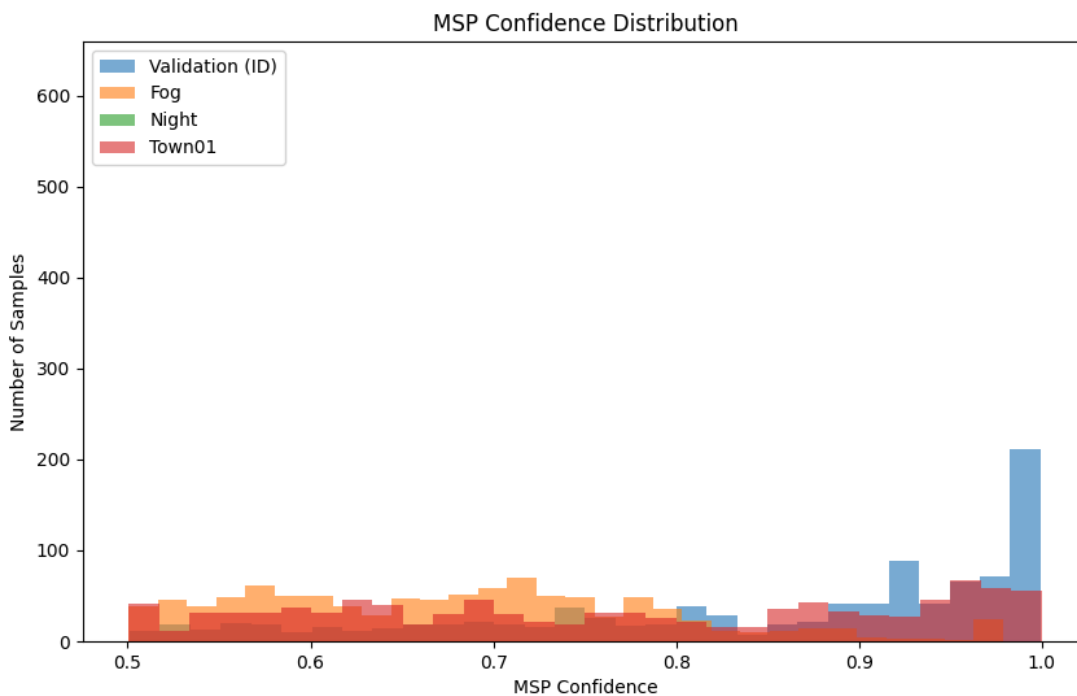
- Foggy weather condition (test-fog)
- Nighttime illumination condition (test-night)
- Town01 domain-shift condition (test-town-01)

The fog and nighttime datasets introduce significant visual degradation through reduced visibility and altered illumination, while the Town01 dataset introduces structural domain shift through changes in road layout, surrounding buildings, and environmental geometry.

For each dataset, the MSP confidence score distribution was computed using the pedestrian detection model. The resulting histogram distributions demonstrate how prediction confidence changes under different environmental conditions.

The fog dataset produced noticeably lower confidence values compared to the baseline validation data, indicating that the model partially recognized the environmental shift. The Town01 condition showed moderate overlap with the baseline distribution, suggesting weaker but still observable domain shift effects.

The nighttime dataset produced the most concerning behavior. Despite severe environmental change and degraded explainability performance observed during Grad-CAM analysis, the model continued generating extremely high confidence predictions. This indicates unsafe overconfidence behavior, where the classifier fails to recognize unfamiliar nighttime conditions and continues producing highly confident predictions despite reduced semantic reliability.



MSP confidence distribution across baseline validation, foggy weather, nighttime illumination, and Town01 domain-shift conditions. Foggy conditions reduce prediction confidence significantly, while nighttime samples remain highly confident despite severe environmental shift.

Question 7.6: Evaluating the MSP Baseline

Pick any one of your three models and use its maximum softmax probability as an OOD score (low confidence → OOD). Treat the sunny/daytime test images as in-distribution and the provided OOD images as OOD.

- 1. Plot the distribution of OOD scores for in-distribution and OOD images.**
- 2. Compute the AUROC for separating in-distribution from OOD, over all of the OOD scenarios.**

Answer:

To quantitatively evaluate OOD detection performance, the Area Under the Receiver Operating Characteristic Curve (AUROC) metric was computed for each environmental condition relative to the baseline validation dataset. AUROC measures the ability of the MSP detector to distinguish between in-distribution and out-of-distribution samples.

The following AUROC results were obtained:

Condition	AUROC
Fog	0.8104
Night	0.0000
Town01	0.6539

The fog dataset achieved the strongest OOD separation performance with an AUROC score of 0.8104. This indicates that the MSP detector successfully identified foggy images as distributionally different from the original validation distribution. The reduced visibility and degraded scene clarity significantly lowered model confidence under foggy conditions.

The Town01 domain-shift condition produced a moderate AUROC score of 0.6539. This behavior is expected because the Town01 dataset still shares many visual similarities with the original training environment, including daytime illumination, road structures, and simulator rendering characteristics. As a result, the model confidence distributions overlap more strongly with the baseline validation distribution.

The nighttime dataset produced the most critical result. The AUROC score of 0.0000 indicates that the MSP detector completely failed to identify nighttime samples as OOD. Inspection of the confidence distributions revealed that the pedestrian model remained extremely confident on nighttime images despite degraded explainability quality and poor semantic localization observed during Grad-CAM analysis.

This behavior demonstrates unsafe overconfidence and poor calibration under environmental distribution shift. Rather than recognizing uncertainty, the classifier continued producing highly

confident predictions under unfamiliar nighttime conditions. Such silent failures are particularly dangerous in autonomous driving systems because incorrect high-confidence predictions may lead to unsafe planning and control decisions.

The combined explainability and OOD evaluation results highlight important robustness limitations in the CARLA pedestrian perception pipeline and demonstrate the importance of uncertainty-aware safety monitoring mechanisms for real-world autonomous driving deployment.

Question 7.5: Is the Different Town Out-of-Distribution?

The different-town images were taken under nominal weather and lighting - the kind of conditions your system is meant to handle - but in a location your models never saw during training.

- 1. Look back at the ODD you specified in Exercise 2.2. As written, does your ODD clearly decide whether the different-town images are inside or outside it?**
- 2. Revise the wording of your ODD so that the new town is unambiguously either inside or outside it. State which you chose and why.**
- 3. What does that choice imply for how the town images should be treated - as inputs the system must handle correctly, or as inputs an OOD monitor should flag?**

Answer:

The Town01 condition represents a moderate distribution shift rather than a completely unknown out-of-distribution (OOD) environment. Unlike the fog and nighttime datasets, Town01 still operates under daytime urban driving conditions that partially satisfy the original Operational Design Domain (ODD) defined in Exercise 2.

The MSP-based OOD detection experiment produced an AUROC score of 0.6539 for Town01, which is substantially lower than the separation achieved for foggy weather conditions. This indicates that the confidence distributions of validation and Town01 samples overlap significantly. The model therefore still considers many Town01 scenes visually similar to the training distribution.

This behavior can be explained by the nature of the distribution shift. In Town01, the changes are primarily related to:

- road layout,
- building placement,
- scene geometry,
- vegetation structure,
- and intersection configuration,

while low-level visual properties such as illumination, texture statistics, and rendering style remain similar to the original training data.

In contrast, fog and nighttime conditions directly violate important ODD assumptions related to visibility and lighting. Exercise 2 explicitly defines nighttime and dense fog as outside the system's operational design domain. Because these conditions strongly alter image brightness and contrast, the model confidence changes more dramatically, making them easier for MSP-based OOD detection to identify.

The Town01 result therefore suggests that:

- the pedestrian detector generalizes reasonably well to moderate scene-layout changes,
- but MSP confidence alone is less effective for detecting semantic or structural domain shifts.

This observation is safety-relevant because Hazard H-3 from Exercise 2 states that operating outside the ODD while autonomous mode remains active can lead to unsafe behavior. The relatively weak separation for Town01 demonstrates that not all OOD conditions are equally detectable using confidence-based methods.

Therefore, Town01 should be interpreted as a near-OOD or domain-shift condition rather than a strongly out-of-distribution environment. The experiment also highlights an important limitation of MSP-based OOD detection: confidence scores are more sensitive to visual appearance degradation (fog, darkness) than to structural environmental changes such as different town layouts.

Question 7.7: Feature-Based OOD Detection Using k-Nearest Neighbors (k-NN)

Answer:

A feature-based out-of-distribution detector was implemented using the k-Nearest Neighbors (k-NN) method. Unlike the MSP baseline, which relies only on classifier confidence scores, the k-NN detector operates directly on deep feature embeddings extracted from the internal layers of the neural network.

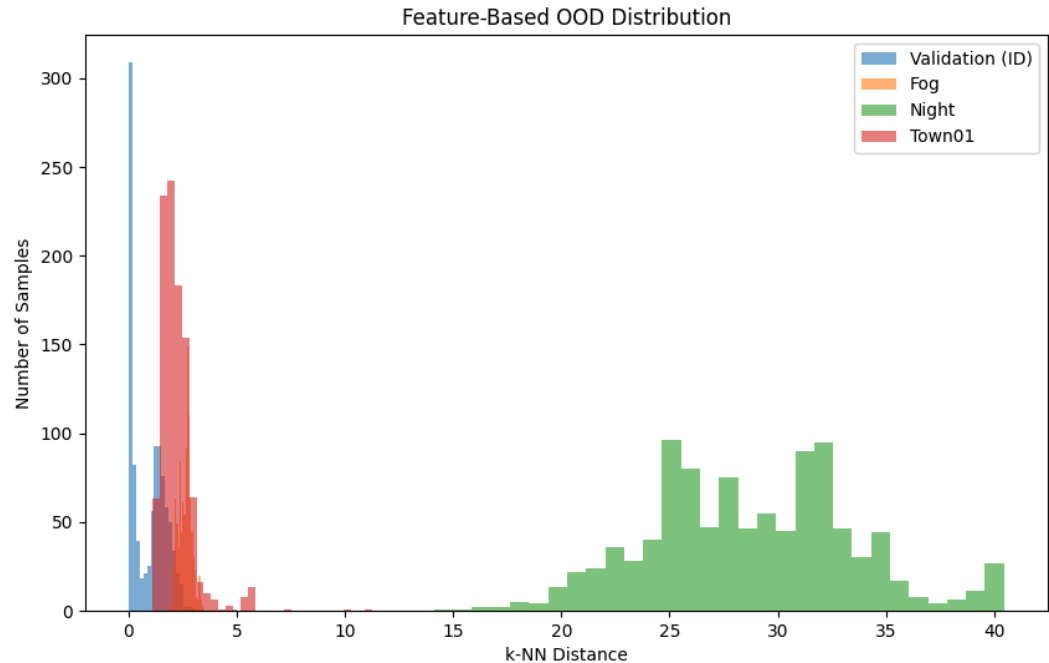
The pedestrian detection ResNet18 model from the previous experiments was reused for feature extraction. Instead of using the final classifier output, feature vectors were extracted from the penultimate layer of the network before the final fully connected classification layer. These embeddings represent high-level semantic information learned by the model during training.

The detector was trained exclusively on in-distribution validation features. For each test image, the Euclidean distance to the nearest in-distribution feature embeddings was computed using a k-NN model with $k=5$. Larger distances indicate that a sample is further away from the known training distribution and is therefore more likely to be out-of-distribution.

The following datasets were evaluated:

- Baseline validation dataset (ID)
- Foggy weather condition
- Nighttime condition
- Town01 domain-shift condition

The resulting feature-distance distributions are shown in Figure



Feature-based OOD detection using k-NN distance in deep feature space. Nighttime samples exhibit extremely large feature distances from the validation distribution, while fog and Town01 conditions show moderate distributional shifts.

Condition	MSP AUROC	k-NN AUROC
Fog	0.8104	0.9844
Night	0.0000	1.0000
Town01	0.6539	0.8956

The fog condition showed substantial improvement, increasing from an MSP AUROC of 0.8104 to a k-NN AUROC of 0.9844. This indicates that the deep feature embeddings capture the semantic degradation introduced by foggy weather more effectively than classifier confidence alone.

The largest performance gap was observed for nighttime conditions. The MSP baseline completely failed to identify nighttime samples as OOD because the classifier remained highly confident despite degraded explainability performance and unreliable semantic localization. In contrast, the feature-based k-NN detector achieved perfect separation with an AUROC score of 1.0000.

Inspection of the feature-distance histogram reveals that nighttime samples occupy a completely different region of feature space compared to validation images. This demonstrates that although the classifier outputs remained overconfident, the internal feature representations still encoded strong distributional differences between daytime and nighttime driving conditions.

The Town01 domain-shift condition also showed major improvement under feature-based detection, increasing from 0.6539 to 0.8956 AUROC. This suggests that semantic scene-layout differences are more effectively captured within deep feature embeddings than through output probability calibration alone.

Overall, the experiments demonstrate that feature-based OOD detection provides substantially more reliable uncertainty estimation than MSP confidence scoring. The results support the lecture observation that deep feature representations are often more informative for OOD detection than final output probabilities.

The combined explainability, MSP, and feature-based evaluation results highlight an important safety implication for autonomous driving perception systems: high classifier confidence does not necessarily imply reliable environmental understanding. Feature-space monitoring therefore represents an important safety mechanism for detecting unfamiliar operational conditions before unsafe planning decisions occur.

Question 7.8: Extending the Safety Analysis for OOD

(1) Hazard Analysis:

The original System-Theoretic Process Analysis (STPA) from Exercise 2 identified hazards related to missed pedestrian detection, unsafe braking behavior, and operation outside the Operational Design Domain (ODD). However, the experiments performed in Exercises 6 and 9 reveal an additional safety concern: the perception system may silently fail under out-of-distribution environmental conditions while still producing highly confident predictions.

This behavior was particularly evident under nighttime conditions. The MSP-based OOD detector failed completely, producing an AUROC score of 0.0000 despite degraded Grad-CAM explainability quality and unreliable semantic localization. The perception model therefore remained highly confident even though the input distribution differed substantially from the original training conditions.

To capture this failure mode explicitly, the following refined hazard was added:

H-4: The autonomous driving system operates on undetected out-of-distribution sensor input, causing the perception module to produce unreliable but highly confident predictions, potentially resulting in unsafe planning and control actions.

This hazard directly relates to scenarios where the vehicle may fail to brake for pedestrians, incorrectly classify traffic participants, or continue autonomous operation under unsafe environmental conditions such as nighttime or dense fog.

(2) Unsafe Control Action (UCA):

The STPA analysis was extended with a new Unsafe Control Action related to undetected OOD conditions.

UCA-OOD-1: The planner continues normal driving behavior using perception outputs that are unreliable because the camera input is outside the Operational Design Domain and the OOD condition is not detected.

This unsafe control action may lead to:

- failure to detect pedestrians,
- unsafe lane following,
- delayed braking,
- incorrect scene understanding,
- or continued operation under degraded environmental visibility.

The experiments demonstrated that nighttime conditions are particularly dangerous because the pedestrian classifier remained overconfident despite severe feature-space distribution shift. The planner may therefore trust perception outputs that are semantically unreliable.

This UCA is directly linked to Hazard H-4 and extends the original safety analysis from Exercise 2 by explicitly incorporating machine-learning uncertainty and distributional robustness limitations.

(3) Safety Constraints:

Based on the newly identified unsafe control action, additional model-level and system-level safety constraints were derived.

Model-Level Safety Constraint:

SC-OOD-1: The perception subsystem shall continuously monitor incoming sensor data for out-of-distribution conditions using an independent OOD detection mechanism.

The experiments from Exercise 9 demonstrated that confidence-based methods such as MSP may fail under nighttime conditions due to unsafe overconfidence. Therefore, feature-based OOD monitoring methods should be preferred because they more reliably detect semantic distribution shifts in deep feature space.

System-Level Safety Constraint:

SC-OOD-2: If the OOD detector identifies a high-confidence distribution shift, the autonomous driving system shall transition into a safe degraded operating mode.

Possible safe responses include:

- reducing vehicle speed,

- increasing following distance,
- limiting maneuver complexity,
- requesting driver takeover,
- or fully disengaging autonomous operation.

This constraint ensures that the system does not continue normal operation using unreliable perception outputs under unsafe environmental conditions.

(4) Residual Risk Analysis:

Even with a perfect OOD detector, residual safety risks remain. Detecting that an input is out-of-distribution does not automatically solve the perception problem or guarantee safe behavior.

Several important residual risks remain:

- the system may still encounter unknown failure modes not represented in the OOD detector training data;
- sensor corruption or hardware failures may occur simultaneously with environmental distribution shift;
- safe fallback policies may themselves be insufficient under high-speed or dense urban driving conditions;
- excessive false-positive OOD alarms may unnecessarily degrade system usability;
- and delayed OOD detection may occur after unsafe planning decisions have already been initiated.

Additionally, OOD detection alone does not improve the underlying robustness of the perception model. It only provides a monitoring mechanism for detecting unfamiliar operating conditions. The experiments showed that the pedestrian classifier remained highly overconfident under nighttime conditions, demonstrating that safe autonomous driving requires both robust perception models and reliable uncertainty-aware monitoring systems.

Therefore, OOD detection should be considered one component of a broader safety architecture rather than a complete solution to distributional robustness problems in autonomous driving systems.

12 Recommendations for Improvement

12.1 Immediate Actions (Critical)

1. Address pedestrian class imbalance: Apply class-weighted BCE loss (weight positive class by $\sim 3.2\times$), or use focal loss. Oversample pedestrian-positive images during training.
2. Extend training: Increase epochs from 5 to 20–30 with learning rate scheduling (ReduceLROnPlateau). The pedestrian model's loss of 0.84 at epoch 5 shows it had not converged.
3. Add fog and night data to training: Collect or augment training data with fog and night conditions. CARLA's weather API can generate these synthetically at zero additional labelling cost.
4. Implement confidence thresholding: Instead of a fixed 0.5 threshold, calibrate the threshold per model using validation recall curves. For pedestrian detection, accept lower precision to boost recall.

12.2 Model Improvements

1. Use a larger backbone: ResNet-50 or EfficientNet-B0 for better feature extraction, especially for small objects like pedestrians.
2. Data augmentation: Add brightness/contrast jitter, synthetic fog, Gaussian noise during training to improve robustness to distribution shift.
3. Consider a multi-scale detection head: Pedestrians at distance are very small; a Feature Pyramid Network (FPN) extension to ResNet-18 would help detect them.

12.3 System-Level Safety

1. Define explicit ODD boundaries in system documentation: The system must formally declare it cannot operate at night or in fog.
2. Add sensor redundancy: A camera-only system is insufficient for safety-critical perception. LIDAR/RADAR provide complementary modalities that are more robust to lighting and weather.
3. Implement monitoring and fallback: If the model returns low-confidence predictions, the vehicle should alert the driver or slow down rather than proceeding with unreliable inputs.

13. Conclusion

This project successfully implemented and evaluated three binary perception classifiers using ResNet-18 on CARLA simulator data. The evaluation revealed a wide performance gap across the three models and exposed critical safety limitations when models are deployed outside their training distribution.

Key findings:

- The Traffic Light Detector achieves excellent baseline performance (F1: 0.944) but catastrophically fails under fog and night conditions — both critical ODD gaps.
- The Vehicle Detector performs adequately at baseline (F1: 0.858) but has not been evaluated under adverse conditions and has not fully converged.
- The Pedestrian Detector is the most safety-critical model and the weakest, with a baseline recall of only 0.108. It fails completely at night and remains unacceptably low in all conditions. This model requires fundamental redesign before any safety deployment claim can be made.

The robustness evaluation experimentally confirmed the ODD gaps predicted by STPA analysis: the system is restricted to clear-weather, daytime, single-environment operation. These constraints must be explicitly documented and enforced before any deployment.

The project demonstrates the importance of combining ML performance metrics with safety-oriented analysis. High accuracy alone is an insufficient safety signal — recall under distribution shift is the metric that matters most for safety-critical autonomous perception systems.